12209095 杜奕衡 HW7

1. (a) $E(\tilde{r}_j) = \alpha_j$

$\sigma_j^2 = Var(\tilde{r}_j) = E[(\tilde{r}_j - E(\tilde{r}_j))^2] = E(\beta_j(\tilde{r}_m - E(\tilde{r}_m)) + \varepsilon_j)^2)$

$= \beta_j^2 E[(\tilde{r}_m - E(\tilde{r}_m))^2] + 2\beta_j(\tilde{r}_m - E(\tilde{r}_m))\varepsilon_j + E(\varepsilon_j^2)$

$= \beta_j^2 \sigma_m^2 + 2\beta_j Cov(\tilde{r}_m, \varepsilon_j) + \sigma_j^2$

$= \beta_j^2 \sigma_m^2 + \sigma_j^2$

(b) $Cov(\tilde{r}_j, \tilde{r}_i) = \sigma_{ij} = E[(\tilde{r}_j - E(\tilde{r}_j))(\tilde{r}_i - E(\tilde{r}_i))]$

$= E[(\beta_j(\tilde{r}_m - E(\tilde{r}_m)) + \varepsilon_j)(\beta_i(\tilde{r}_m - E(\tilde{r}_m)) + \varepsilon_i))$

$= \beta_j\beta_i\sigma_m^2 + \beta_j(Cov(\tilde{r}_m, \varepsilon_i) + \beta_i Cov(\tilde{r}_m, \varepsilon_j) + Cov(\varepsilon_j, \varepsilon_i)$

$= \beta_j\beta_i\sigma_m^2$

2. (a) if $\beta_{wm} = \beta_{wv} = 0$, $E(\tilde{r}_w^1) = r_f + \beta_{wm}(E(\tilde{r}_m) - r_f) + \beta_{wv} E\tilde{r}_v = r_f$

(b) $E(\tilde{r}_w^2) = r_f + E(\tilde{r}_m) - r_f = E(\tilde{r}_m)$

$E(\tilde{r}_w^3) = r_f + E(\tilde{r}_v) - r_f = E(\tilde{r}_v)$

(c) from (a) and (b), we could long β_{wm} $\tilde{r}_w^2$, β_{wv} $\tilde{r}_w^3$, short $(\beta_{wm} + \beta_{wv} - 1) r_f$ and 1 $\tilde{r}_w^4$

$E(\beta_{wm}\tilde{r}_w^2 + \beta_{wv}\tilde{r}_w^3 - (\beta_{wm} + \beta_{wv} - 1)\tilde{r}_w - \tilde{r}_w^4)$

$= \beta_{wm} E(\tilde{r}_m) + \beta_{wv}(E(\tilde{r}_v)) - \beta_{wm} r_f - \beta_{wv} r_f + r_f - r_f - \beta_{wm}(E(\tilde{r}_m) - r_f)$

$- \beta_{wv}(E(\tilde{r}_v) - r_f) - \Delta$

$= -\Delta$

for the unit of asset we long and short is the same, we need no money down but to gain a positive return $-\Delta$.

3. (a)

CAPM	APT	Method
competitive Equilibrium	No arbitrage	investors
M-V investors	All investors who prefer more to less	risk factors
$\tilde{r}_m$ (result)	$\tilde{r}_{RF}$ (Assump)	assets
any asset	well-diversified asset	conclusion
$E(\tilde{r}_j) = r_f + \beta_j[E(\tilde{r}_m) - r_f]$	$E(\tilde{r}_w) = r_f + \beta_w (E(\tilde{r}_{RF} - r_f))$	

(b) After gained the expected return r, current price can be derived by:
$r = \frac{C}{P} - 1 \Rightarrow P = \frac{C}{1+r}$ (C is future payoff)

(c) They are not compatible, as their key assumptions are different.
However, if we take all stronger assumptions they make.
namely: competitive Equilibrium ; M-V investors ; $\tilde{r}_{RF} = \tilde{r}_m$;
well-diversified asset.
CAPM and APT will be identical.